

Fri - 04-11-2025, Solid rockets (Example and design)
(Sec. 4)

Continue at previous example, we obtained $r)_{20}, r)_{70}$ at $T=40^\circ\text{C}$ in FL, now, find n :

$$n = \frac{\ln\left(\frac{r'_2}{r_2}\right)}{\ln\left(\frac{P_{02}}{P_{01}}\right)} = 0.9698 < 1 \quad \text{stable @ } T=40^\circ\text{C in FL}$$

Maximum temp. for stable operation in FL?

use sensitivity Eqⁿ:

$$\text{@ } 20 \text{ atm} \quad 0.005 = \frac{2}{r)_{20} + 3} \frac{r)_{20} - 3}{T_{\max} - (-10)} \quad (1)$$

$r)_{20}$ is r at 20 atm @ $T_{\max}$

$$\text{@ } 70 \text{ atm} \quad 0.0095 = \frac{2}{r)_{70} + 8} \frac{r)_{70} - 8}{T_{\max} - (-10)} \quad (2)$$

$r)_{70}$ is r at 70 atm @ $T_{\max}$

Two Eqⁿ, 3 unknowns $\begin{cases} r)_{20} \\ r)_{70} \end{cases} \quad T_{\max}$

$$\text{Divide (2) to (1)} \implies 1.4 = \frac{r)_{70} - 8}{r)_{70} + 8} \times \frac{1}{\left(\frac{r)_{20} - 3}{r)_{20} + 3}\right)}$$

$T_{\max}$ corresponds $n=1$ (border of stability)

(P-1)

$$\begin{aligned}
 r_{20} &= a(20)^1 \\
 r_{70} &= a(70)^1
 \end{aligned}
 \implies \frac{r_{20}}{r_{70}} = \frac{20}{70} \implies \boxed{r_{20} \times 3.5 = r_{70}} \quad (4)$$

put (4) in (3) $\implies 1.4 = \frac{3.5 r_{20} - 8}{3.5 r_{20} + 8} \times \frac{1}{\frac{r_{20} - 3}{r_{20} + 3}}$

Solve $\rightarrow r_{20} = 4.01 \frac{\text{mm}}{\text{s}}$, $r_{70} = 3.5 \times 4.01 = 14.035 \frac{\text{mm}}{\text{s}}$

These are burn rates at $T_{\max}$

Use: $0.005 = \frac{2}{4.01 + 3} \frac{4.01 - 3}{T_{\max} - (-10)} \implies \boxed{T_{\max} = 47.63^\circ \text{C}}$

Now, consider a design problem of 2-stage solid rocket.

The rocket is similarly-staged with $\epsilon = \lambda = 0.15$, $M_L = 6000 \text{ kg}$

Propellant $\left\{ \begin{array}{l} \text{Fuel: } 36\% \\ \quad \left\{ \begin{array}{l} 21\% \text{ AL(cr)} \\ 15\% \text{ HTPB (C}_4\text{H}_6 \text{ butadiene)} \end{array} \right. \\ \text{ox: } 64\%, \text{ AP: } \text{NH}_4\text{ClO}_4 \text{ (I)} \end{array} \right.$

$P_o = 94 \text{ atm}$, $\frac{A_e}{A^*} = 22$, $\rho_p = 1940 \frac{\text{kg}}{\text{m}^3}$, $T_p = 298.15$

a) Run CEA, find T_o , c^* , C_F (thrust coeff.), γ , c_p ,
mole fraction of species. (P-2)

Solution: Before going to CEA, compute fuel percentages:

we have 36% fuel $\left\{ \begin{array}{l} 21\% \text{ AL(cr)} \rightarrow \text{AL(cr)}\% : \frac{21}{36} = 58.33\% \\ 15\% \text{ HTPB} \rightarrow \text{HTPB}\% : \frac{15}{36} = 41.67\% \end{array} \right.$

$$r = O/F = \frac{\text{mass of ox}}{\text{mass of fuel}} = \frac{64}{36} = 1.77$$

G₀ to CEA:

$$\boxed{\begin{array}{l} I_{sp} \times g_e = u_{eq} = 2826 \frac{m}{s} \\ CF = c_r = 1.7689 \end{array}}$$

b) compute Δu_b , M_o , M_s , M_p for both stages.

Solution: For similarly staged rockets:

$$\Delta u_b = n \underbrace{g_e I_{sp}}_{u_{eq} \text{ (from CEA)}} \ln \left(\frac{1+\lambda}{\epsilon+\lambda} \right)$$

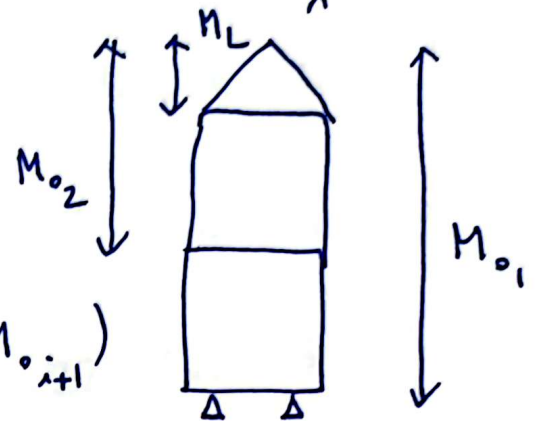
number of stages $\rightarrow \boxed{\Delta u_b = 7600 \frac{m}{s}}$

For similarly-staged rockets, we have:

$$\frac{M_{o,i}}{M_L} = \prod_{i=1}^n \frac{1+\lambda}{\lambda} \Rightarrow \frac{M_{o,i}}{M_L} = \left(\frac{1+\lambda}{\lambda} \right)^2$$
$$\Rightarrow \boxed{M_{o,i} = 354201 \text{ kg}}$$

$$d \quad \lambda_i = \frac{M_{o,i+1}}{M_{o,i} - M_{o,i+1}} \Rightarrow \begin{cases} \lambda_i = 0.15 \\ i=1 \end{cases} \quad M_{o,2} = \frac{M_{o,1}}{\frac{1}{\lambda} + 1}$$

$$\Rightarrow \boxed{M_{o,2} = 46100 \text{ kg}}$$



$$\varepsilon_i = \frac{M_{s,i}}{M_{o,i} - M_{o,i+1}} \Rightarrow M_{s,i} = \varepsilon_i (M_{o,i} - M_{o,i+1})$$

$$M_{s,1} = \varepsilon (M_{o,1} - M_{o,2}) = 46215 \text{ kg}$$

$$M_{s,2} = \varepsilon (M_{o,2} - M_L) = 6015 \text{ kg}$$

$$M_{o,i} = M_{s,i} + M_{p,i} + \underbrace{M_L}_{M_{o,i+1}}$$

$$\begin{cases} M_{p,1} = M_{o,1} - M_{o,2} - M_{s,1} = 261886 \text{ kg} \\ M_{p,2} = M_{o,2} - M_L - M_{s,2} = 34085 \text{ kg} \end{cases}$$

c) Find propellant volume for both stages:

$$\left. \begin{array}{l} \rho_p = 1940 \frac{\text{kg}}{\text{m}^3} \\ M_{p,1} = 261886 \text{ kg} \\ M_{p,2} = 34085 \text{ kg} \end{array} \right\} \Rightarrow V_p = \frac{M_p}{\rho_p} \Rightarrow \begin{cases} V_{p,1} = 134.99 \text{ m}^3 \\ V_{p,2} = 17.57 \text{ m}^3 \end{cases}$$

d) Determine A^* for the first stage such that $T = 2.9 W$

Solution: From CEA, read CF (which is thrust coeff.)

$$CF = C_F = \frac{\tau}{P_o A^*} \Rightarrow \tau = C_F P_o A^* = 2.9 W = 2.9 M_o g_e$$

$$\Rightarrow A^* = \frac{2.9 M_o g_e}{C_F P_o} = 0.5979 \text{ m}^2 \quad A_e = 22 A^* = 13.154 \text{ m}^2$$

e) Assume the propellant for each stage is a solid cylinder that is the same diameter as the first stage nozzle exit, find the height of the propellant in each stage.

Solution:

$$A_e = \pi \frac{D_e^2}{4} \Rightarrow D_e = \sqrt{\frac{4 A_e}{\pi}}$$
$$L_1 = \frac{Vol_1}{A_e} \quad L_2 = \frac{Vol_2}{A_e}$$